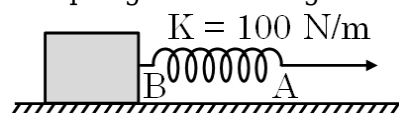


PART-1 : PHYSICS

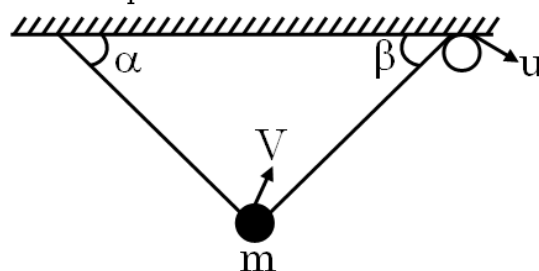
SECTION-I (i)

1) A block lying on a smooth surface with spring connected to it is pulled by an external force as shown. The velocity of ends A and B of the spring are 4 m/s and 2 m/s respectively. If the energy of the spring is increasing at the rate of 20 J/sec, then the stretch in the spring is :-



- (A) 1.0 cm
- (B) 2.0 cm
- (C) 10 cm
- (D) 20 cm

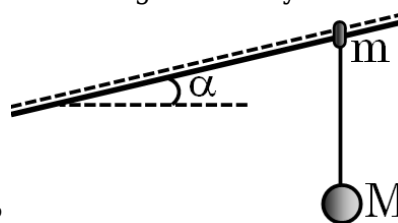
2) Ignoring the gap between pulley & ceiling find the speed V of particle of mass ' m ' fixed on the



string. String is pulled with speed u as shown in figure.

- (A) $V = \frac{u}{\sin(\alpha + \beta)}$
- (B) $V = \frac{u}{\cos(\alpha + \beta)}$
- (C) $V = u \sin(\alpha + \beta)$
- (D) $V = u \cos(\alpha + \beta)$

3) A smooth rod is fixed at an angle α with respect to the horizontal. A little ring of mass m can slide along the rod, to which a long thread is attached. A small sphere of mass M is attached to the thread. Initially the ring is held motionless, and the thread hangs vertically. What is the acceleration

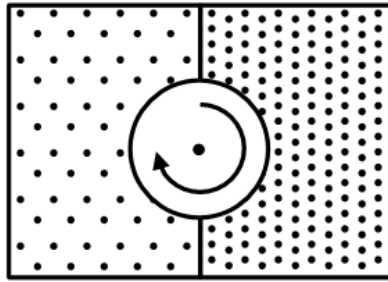


of the sphere immediately after the ring is released ?

- (A) $\frac{(m + M)g \sin \alpha}{m + M \sin^2 \alpha}$

- (B) $\frac{(m + M)g \sin^2 \alpha}{m + M \sin^2 \alpha}$
 (C) $\frac{(m + M)g \sin \alpha \cos \alpha}{m + M \sin^2 \alpha}$
 (D) $\frac{(m + M)g \sin \alpha}{M + m \sin^2 \alpha}$

4) A homogeneous disk was spun around its axis to an angular velocity of ω and placed on the boundary between two horizontal half-planes so that its center was exactly on the boundary (see figure; top view). The coefficient of friction between the disk and one half-plane is k , and between the disk and the other half-plane is $2k$. Find the acceleration of the center of the disk immediately



after it landed on the surface.

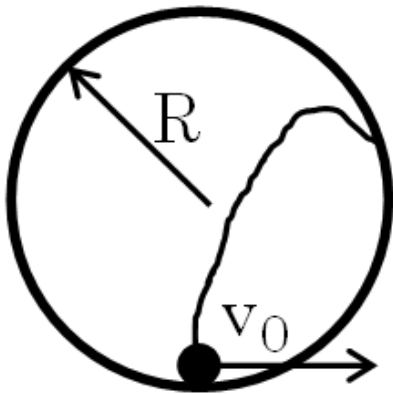
- (A) $\frac{kg}{\pi}$
 (B) $\frac{2kg}{\pi}$
 (C) $\frac{kg}{2\pi}$
 (D) $\frac{4kg}{\pi}$

SECTION-I (ii)

1) To study the variation of velocity in some motion, the velocity vectors at different times in the given motion are drawn from the same initial point, with their lengths proportional to the magnitude. In which of the following cases the curve joining the final ends of the velocity vectors is a circle?

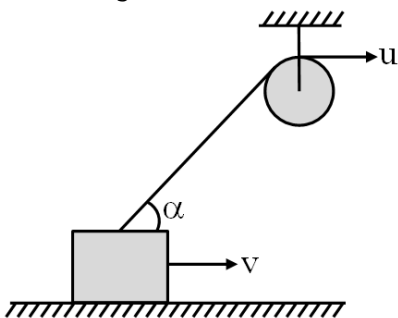
- (A) Projectile motion in uniform gravity.
 (B) Motion of a particle on a wheel rolling uniformly on a horizontal surface.
 (C) Motion of a particle falling along a smooth brachistochrone (cycloidal curve with least time of fall).
 (D) Planetary motion in an elliptical orbit around the Sun.

2) In a smooth ring-shaped trough located in a vertical plane there is a small ball. The ball, which is in a position of equilibrium, is given such a horizontal velocity that after breaking away from the trough at a certain point it falls onto the trough at the starting point (see Fig.).



- (A) The initial velocity v_0 is $\sqrt{\left(\frac{7Rg}{2}\right)}$.
- (B) The velocity at the instant the ball breaks away from the trough is $\sqrt{\left(\frac{Rg}{2}\right)}$.
- (C) The maximum height reached by the ball from the initial position is $\frac{3\sqrt{3}R}{4}$.
- (D) The angle made by velocity vector with vertical when breaking away from the trough is 60° .

3) A weightless, inextensible thread thrown over a pulley is attached to a load of mass m located on a smooth horizontal surface. A boy pulls the horizontal end of the thread with a constant speed u . At some instant, the inclined section of the thread makes an angle α with the horizontal. The pulley is at a height h , the dimensions of the block and the pulley can be neglected. The acceleration of gravity is g . Consider only the motion of the block before it lifts off the surface. The middle point of the string between the block and the pulley (not a fixed point on the string) is P_0

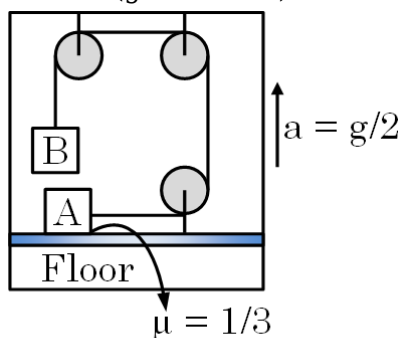


- (A) 'P' is moving horizontally with speed $\left(\frac{v}{2}\right)$.
- (B) The tension in the string at the given instant is $T = \frac{mv^2 \sin^2 \alpha}{h \cos \alpha}$.
- (C) The magnitude of acceleration of the block is increasing.
- (D) The normal acceleration of the block is zero.

SECTION-I (iii)

1) Consider the given system in which two blocks connected by ideal string and pulley system placed in lift, which is moving upward with an acceleration of $\frac{g}{2}$ m/s² as shown in figure. Mass of blocks A &

B are 0.1 kg each & friction coefficient between floor & block A is $\mu = \frac{1}{3}$. Then match the list I against list II. ($g = 10 \text{ m/s}^2$). All units are in S.I. All are at rest at $t = 0$. (All strings and pulleys are



ideal) $m = 1/10 \text{ kg}$

	List-I	List-II	
(P)	Magnitude of acceleration of each block with respect to lift	(1)	5
(Q)	Tension in string	(2)	2
(R)	Work done by tension on block A from ground reference frame in time $2/\sqrt{5}$ sec	(3)	0
(S)	Work done by tension on block B from ground frame of reference	(4)	1
		(5)	4

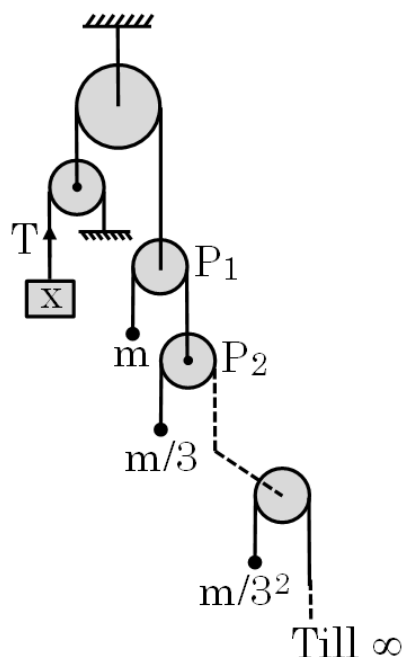
(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 3$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 4$

2) Assume all pulleys to be massless and smooth. Match the following. ('x' represents unknown



mass) [Take $g = 10 \text{ m/s}^2$]

For More Join: @IITJEE_Advanced

	List-I		List-II
(P)	If $x = \frac{m}{2}$	(1)	Acceleration of pulley P_1 is 2 m/s^2 .
(Q)	If $x = m$	(2)	Tension $T = \frac{mg}{2}$
(R)	If $x = \frac{m}{4}$	(3)	Acceleration of pulley P_1 is 2.5 m/s^2 .
(S)	If $x = 2m$	(4)	Tension $T = \frac{2mg}{3}$
		(5)	Acceleration of pulley P_1 is 8 m/s^2 .

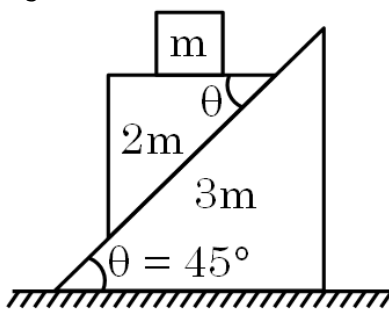
(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$

(D) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$

3) Three masses are kept as shown in figure. Assume all surfaces to be smooth. Just after system is



released from rest find the following.

	List-I		List-II
(P)	Acceleration of mass m is	(1)	$\frac{2}{7}g$
(Q)	Acceleration of mass $2m$ in horizontal direction	(2)	$\frac{5}{7}g$
(R)	Acceleration of mass $2m$ is	(3)	$\frac{3}{7}g$
(S)	Acceleration of mass $3m$ is	(4)	$\frac{\sqrt{34}}{7}g$
		(5)	Zero

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

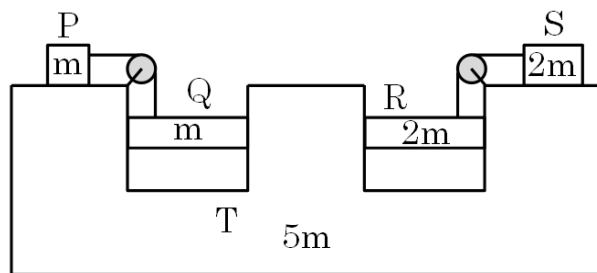
(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

4) P, Q, R, S are four blocks, T is a big block supporting all of the smaller blocks as shown in the figure. Assume all pulleys to be ideal & surfaces smooth. If system is released from rest then answer

For More Join: @IITJEE_Advanced



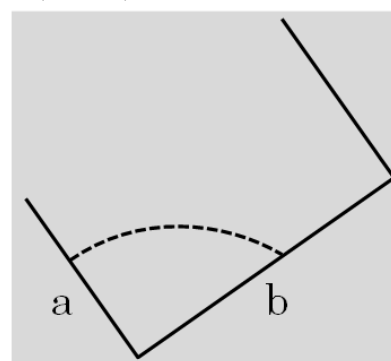
the following.

	List-I		List-II
(P)	Acceleration block P w.r.t. block T is	(1)	$\frac{g}{19}$
(Q)	Acceleration of block T is	(2)	$\frac{9g}{19}$
(R)	Acceleration of block S w.r.t. T is	(3)	g
(S)	Relative acceleration of block P w.r.t. S is	(4)	$\frac{10g}{19}$
		(5)	$\frac{g}{48}$

- (A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$
 (B) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$
 (C) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 2$
 (D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

SECTION-II

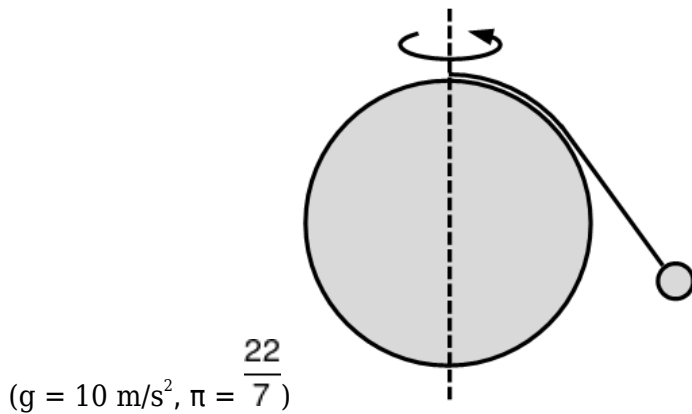
1) In a rectangular box, a ball bounces elasticity back and forth along the same trajectory, hitting the left wall and the bottom at points a and b from the bottom corner of the box. The distance of a from the common corner is 3m and that of b is 4m. What is the time (in sec) between successive



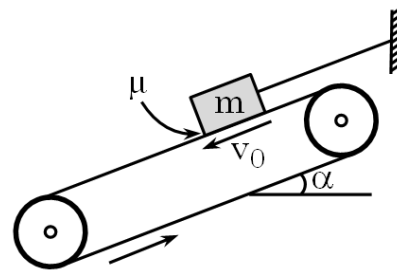
impacts of the ball? The acceleration due to gravity is $g = 10\text{m/s}^2$.

2) The ball rotates with a frequency $\nu = 0.7 \text{ s}^{-1}$ around a vertical axis passing through its centre. A thread with a small body is attached to the top point of the ball (see figure). The length of the thread is equal to a quarter of the circumference of the large circle of the ball. $\frac{2}{3}$ of the length of the

thread is in contact with the surface of the ball. The radius of the ball is $\frac{Ng}{\pi^2 \nu^2 (18 + \pi\sqrt{3})}$. Find N.

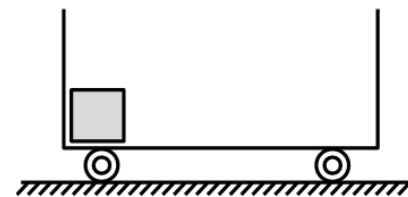


3) On a conveyor belt moving at a constant speed v_0 downwards at an angle $\alpha = 37^\circ$ to the horizontal, there is a brick, held from moving downwards together with the belt by means of a cord tied to it, fixed at the top (see fig.). The cord is cut. What work (magnitude in J) will the friction force acting on the brick do by the time the brick reaches the speed of the conveyor, and has a kinetic



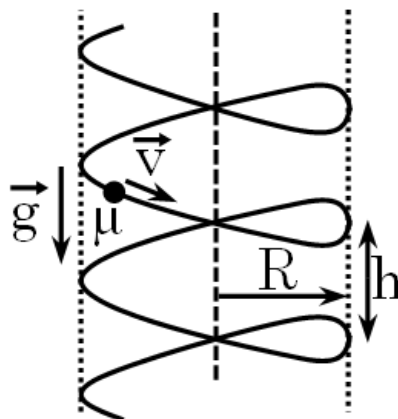
energy of 8J at that instant? Coefficient of friction $\mu = 0.25$.

4) At the left edge of a cart of length $L = 0.25 \text{ m}$ and mass $M = 1 \text{ kg}$ lies a cube of mass $m = 0.25 \text{ kg}$ (see fig.). The cube is given a horizontal velocity $v_0 = 1 \text{ m/s}$ to the right by pushing it, while the cart is initially stationary. Determine at what distance from the left edge of the cart the cube will be after it stops sliding relative to the cart. The coefficient of friction of the cube on the bottom of the cart is $\mu = 0.1$. The impacts of the cube on the walls are considered to be absolutely elastic. The cart moves



along the table without friction. If the answer is $5x \text{ cm}$. Mark x .

5) A long wire is wound into a spiral of radius R with a pitch h (Fig.). The axis of the spiral is located vertically. A bead slides along the spiral; the coefficient of friction between the wire and the bead is μ . Find the square of steady-state speed v^2 of the bead. (Given $h/R = 5\pi/2$ and $\mu = 3/4$. Also, square



of radius R is $(9/164) m^2$, $g = 10 m/s^2$

6) If a spring block system is kept on a rough horizontal surface with one end of spring fixed. The block could stay in equilibrium up to a maximum elongation of e . If the block is pulled by $5.5e$ and released, the total work done by friction till the block eventually stops is $5Nke^2$, where k is the spring constant. Find the nearest value of integer N .

PART-2 : CHEMISTRY

SECTION-I (i)

1) Identify which of the following species have a linear triatomic ion :-

- (I) Calcium cyanamide
- (II) Magnesium carbide
- (III) Product of reaction between ICl_4 & $CsCl$
- (IV) Product of lewis acid-base interaction between $CClF_3$ & AsF_5

- (A) I, II, III only
- (B) I, II only
- (C) I & III only
- (D) I, II, III and IV

2) Select the triad in which all compounds/species are isoelectronic.

- (A) H_2BNH_2 , C_2H_6 and B_2H_6
- (B) $H_3N \cdot BH_3$, B_2H_6 and C_2H_6
- (C) B_2H_6 , C_2H_6 and $N_2H_6^{2+}$
- (D) $H_3N \cdot BH_3$, C_2H_6 and $N_2H_6^{2+}$

3) Which of the following species behave as paramagnetic diradical.

- (A) Cl_2 , S_2
- (B) ClO_2 , NO_2
- (C) S_2 , O_2

(D) ClO_2 , O_2

4) Consider the following salts -

(P) NH_4NO_2 , (Q) NH_4NO_3

(R) NH_4N_3 (S) $\text{N}_2\text{H}_5\text{NO}_3$

(T) $\text{N}_2\text{H}_5\text{NO}_2$ (U) $(\text{NH}_4)_2[\text{Ce}(\text{NO}_3)_6]$

Select the **INCORRECT** option.

- (A) Except one, all the salt have -ve oxidation state of nitrogen in cation and +ve oxidation state of nitrogen in anion.
- (B) All the salts have at least one Nitrogen in tetrahedral geometry.
- (C) All the salts have at least one nitrogen in triangular planar geometry.
- (D) Average oxidation number of nitrogen is lowest in (R) & highest in (U)

SECTION-I (ii)

1) For second period homonuclear diatomic molecules, both 2s and 2p orbitals are used in the formation of molecular orbitals. The shape of the molecular orbitals are identified by linear combination of atomic orbitals, followed by normalisation of wavefunction and then taking the square of the wavefunction.

Select the **CORRECT** statement(s) about homonuclear diatomic molecules of 2nd period.

- (A) Six 2p atomic orbitals form a σ_b and σ^* pair and two doubly degenerated π_b & π^* molecular orbitals are present.
- (B) σ bonding molecular orbitals are characterised by the absence of nodal plane perpendicular to internuclear axis between the two nuclei.
- (C) π -bonding molecular orbital have a nodal plane along bond axis and another nodal plane perpendicular to the nodal plane between the two nuclei.
- (D) The order of bond length & bond strength is just reverse in N_2 , C_2 and F_2

2) When halogen reacts with hydrogen they produces hydrogen halide, hydrogen halide dissolve in water to produce hydrohalic acid.

Select which is/are **CORRECT** for hydrogen halides / hydrohalic acid.

- (A) Hydrogen halides are colorless diatomic gases at room temperature but HF is hydrogen bonded oligomer low boiling colorless liquid (293 K).
- (B) HCl gas have lowest melting point & boiling point, and reaction of HCl_{aq} with iron proceeds to FeCl_3 & H_2
- (C) The ΔH_f° of gaseous HX diminish rapidly with increasing molecular weight and for HI it is endothermic
- (D) The thermal stability and boiling point change regularly on moving from Fluorine to Iodine.

3) Select statement(s) which is/are **CORRECT** for orthoboric acid.

- (A) It can be prepared by acidifying the aqueous solution of borax.
- (B) It can be prepared by hydrolysis of boron halides and boron hydrides.
- (C) In aq. solution with polyhydroxy alcohols its acidic strength is explained by increased pKa.

For More Join: @IITJEE_Advanced

(D) In aq. NaOH it is act as H^+ donor acid.

SECTION-I (iii)

1) Surface tension of liquids is depends on the magnitude of intermolecular forces.

Match List-I and List-II

List-I (Liquid)		List-II (Surface tension in J/m^2 and Dominant intermolecular forces)	
(P)	Cyclohexanol	(1)	0.0477 , Hydrogen bond
(Q)	Diethyl ether	(2)	0.425 , London dispersion
(R)	n-Hexane	(3)	0.0170 , Dipole-dipole
(S)	Ethylene glycol	(4)	0.0344 , Hydrogen bond
		(5)	0.0184 , London dispersion

(A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

(B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$

(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 1$

2) Following question is based on inorganic reactions (List-I) and the result(s) of reactions (List-II)

Find the correct code of your answer.

List-I (Reaction)		List-II (Results)	
(P)	Chromium metal + conc. HNO_3	(1)	Highly poisonous gas produces which is explodes in contact with Cl_2
(Q)	$H_3PO_3 + PCl_3$	(2)	Yellowish white powder which is moist in air
(R)	Pt with aqua regia	(3)	Deep red fumes with clear solution
(S)	Heating of phosphinic acid	(4)	Dibasic oxoacid is produces
		(5)	Passive film of oxide on the surface

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 1$

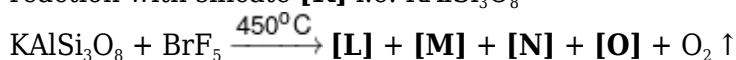
(B) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

3) Use given information to solve given matching type question and find correct code of your answer.

The vigorous fluorinating activity of an interhalogen compound [X] i.e. BrF_5 is demonstrated by reaction with silicate [K] i.e. $KAlSi_3O_8$



If [L] , [M] , [N] and [O] all are binary fluoride compounds and atomic number of atom in fluorides is changed as $L > M > N > O$

For More Join: @IITJEE_Advanced

List-I (Binary fluoride compound)		List-II (Characteristics / Specification of Binary fluoride)	
(P)	[L]	(1)	Can produce hexafluoro anion on reacting with water
(Q)	[M]	(2)	High melting point and insoluble in water
(R)	[N]	(3)	Covalent molecule with square pyramidal shape
(S)	[O]	(4)	Electrolyte in Moissan process
		(5)	Yellow green liq. at room temperature

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 2$

(D) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

4) Match the substance (List-I) with their attributes (List-II).

Find the correct code of your answer.

List-I (Substance)		List-II	
(P)	TeO_2	(1)	Used in nuclear fuel industry for enrichment of Uranium
(Q)	PH_3 (Phosphine)	(2)	Oxidizing agent
(R)	ClF_3	(3)	Toxic gas
(S)	Volatile product of Kipp's apparatus	(4)	Hybridisation of central atom is sp^3
		(5)	Precipitated by $\text{CuSO}_4(\text{aq.})$

(A) $P \rightarrow 2,3; Q \rightarrow 3,4,5; R \rightarrow 1; S \rightarrow 3,5$

(B) $P \rightarrow 2; Q \rightarrow 3,5; R \rightarrow 1,2; S \rightarrow 3,5$

(C) $P \rightarrow 2,3,4; Q \rightarrow 2,3,4; R \rightarrow 1,2,4; S \rightarrow 3,5$

(D) $P \rightarrow 3,4; Q \rightarrow 3,4,5; R \rightarrow 1,3; S \rightarrow 3,4,5$

SECTION-II

1) Consider the following reaction

• Hydrazine + $\text{I}_2 \rightarrow \text{Gas-1} + \text{Gas-2}$

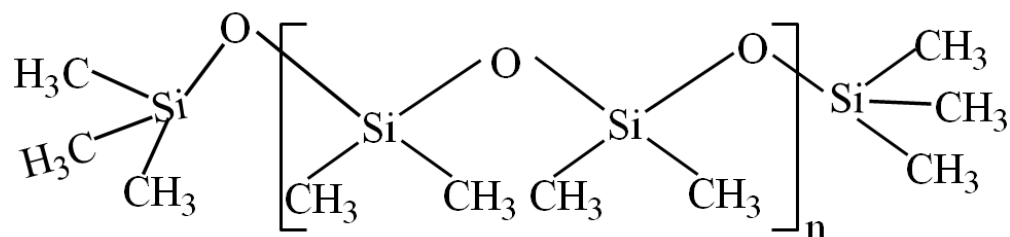
• $\text{MnO}(\text{OH})_2 + \text{HCl}_{(\text{aq.})} \xrightarrow{\Delta} \text{Gas-3} + \text{other products}$

For More Join: @IITJEE_Advanced

- Galena + $O_3 \rightarrow$ **Gas-4** + other products
- Hydrolith_(fused) $\xrightarrow{\text{Electrolysis}}$ **Gas-5** + other products
- Water gas preparation $\rightarrow$ **Gas-5** + **Gas-6**

Identify all the gases (1 to 6) and calculate the sum of bond order(s) in **(Gas-1) + (Gas-2) + (Gas-3) + (Gas-4) + (Gas-5) + (Gas-6)**

2) Given silicone molecule (where $n = 25$) is prepared by hydrolysis & condensation of methyl substituted chlorosilanes.



Let :-

P = Number of molecules of dimethyl chlorosilanes required

Q = Number of molecules of trimethyl chlorosilanes required

R = Total number of HCl molecules formed during reaction.

S = Total number of Si-O-Si bonds in given molecule.

Find the value of $P + Q + R + S$?

3) Total number of paramagnetic species from given below

O_2 , O_3 , KO_3 , KO_2 , Cl_2^- , NO_2^+ , ClO_2 , ClO_2^- , S_2 , B_2H_6 , N_2O_4 , N_2O , NO , P_4 , Xenon

4) An impure white solid sample which contain Calcium chloride , Calcium hypochlorite, Calcium hydroxide & some water of crystalization. That impure solid is widely used as an oxidizing agent in Textile mills, Looms , Hosiery laundry and Organic synthesis.

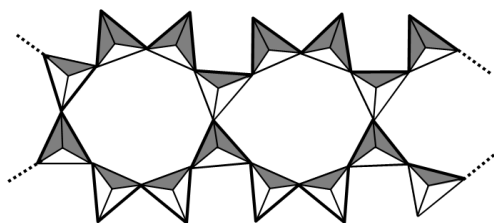
Find the number of **CORRECT** statement(s) for the given impure solid sample.

- Have chlorine with oxidation number = +1
- Have chlorine with oxidation number = -1
- Produce red color on Bunsen burner flame
- The oxo anion of given salt produces ClO_3^- & Cl^- on disproportionation
- Produces I_2 when react with KI in acidic solution
- In Moist air produces pungent smelling gas
- It can decolorize moist litmus paper
- It is used to disinfects water
- ClO_2 is the anhydride of that oxyacid whose anion present in given salt

5) The rare 3T double chain silicate occurs in Xonolite $[Ca_6Si_xO_y(OH)_2]$

Calculate the sum of x and y.

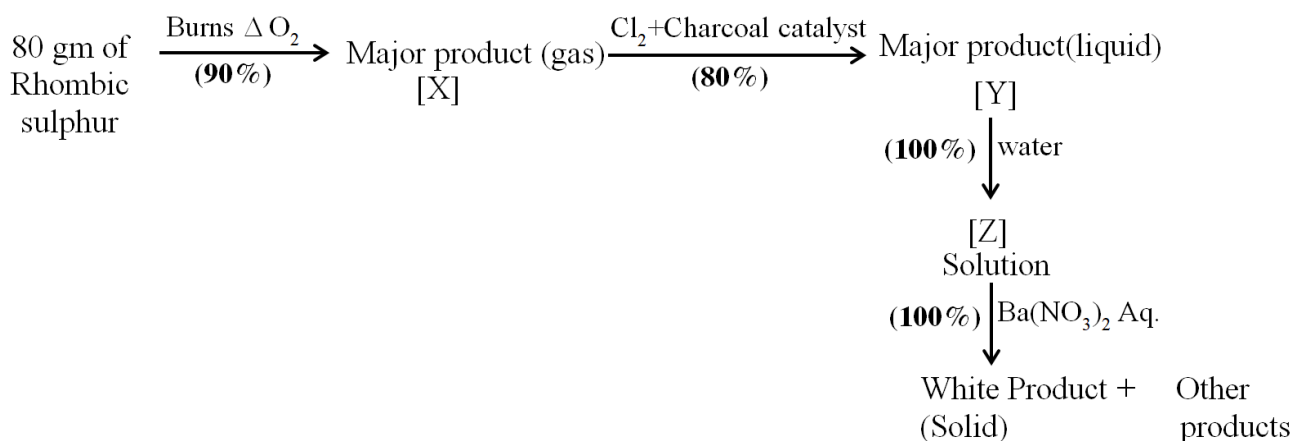
For More Join: @IITJEE_Advanced



(A portion of the Silicate part of Xonolite)

6) If the reaction sequence given below is carried out with 80 gm of Rhombic sulphur, the amount of white solid formed finally (in gm) is.

(Ignore all the values after decimal place in final answer) :-



The % yield of [X] , [Y] , [Z] and the final product white solid is given in parentheses

[Atomic mass of H = 1 , C = 12 , O = 16 , Cl = 35.5 , S = 32 , Ba = 137 , N = 14]

PART-3 : MATHEMATICS

SECTION-I (i)

1) Number of real solution(s) of the equation $2^{|x|} (\tan^{-1} |x| + 1) = \frac{4}{3}$, is

- (A) 0
- (B) 1
- (C) 2
- (D) 4

2) Let a relation R on the interval $\left[0, \frac{\pi}{2}\right)$ is defined as $x R y$ if and only if $\sec^2 x - \tan^2 y = 1$. Then R is :

- (A) an equivalence relation
- (B) both reflexive and transitive but not symmetric
- (C) both reflexive and symmetric but not transitive
- (D) reflexive but neither symmetric not transitive

3) If $\sin\alpha \cos\beta = -\frac{1}{2}$, then $\cos\alpha \cdot \sin\beta$ can take which of the following values

- (A) $\frac{3}{4}$
- (B) $\frac{1}{4}$
- (C) $\frac{\sqrt{3}}{2}$
- (D) $-\frac{\sqrt{3}}{2}$

4) The smallest positive root of the equation $\sqrt{\sin(1-x)} = \sqrt{\cos x}$ is equal to -

- (A) $\frac{1}{2} + \frac{3\pi}{4}$
- (B) $\frac{1}{2} + \frac{7\pi}{4}$
- (C) $\frac{1}{2} + \frac{11\pi}{4}$
- (D) $\frac{1}{2}$

SECTION-I (ii)

1) Let $f(x) = \sin(\cos^{-1}(\tan(\cos^{-1}x)))$. If α be a real solutions of the equation $f(f(x)) = f(x)$, then the value of $\left[\frac{\alpha}{2}\right]$ can be -
(where $[.]$ is greatest integer function)

- (A) 1
- (B) 0
- (C) -1
- (D) 2

2) Let $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be a function such that $g(1,1) = 2$;
 $g(\alpha + 1, \beta) = g(\alpha, \beta) + \alpha \quad \forall \alpha, \beta \in \mathbb{N}$ and
 $g(\alpha, \beta + 1) = g(\alpha, \beta) - \beta \quad \forall \alpha, \beta \in \mathbb{N}$
Now, if for some $x, y \in \mathbb{N}$; $g(x, y) = 2001$
then which of the statement(s) is/are true ?

- (A) number of ordered pairs (x, y) is 2
- (B) maximum value of $(x + y)$ is 3999
- (C) minimum value of $(x + y)$ is 2000
- (D) minimum value of $(x + y)$ is 1999

3) Let $a = \frac{\pi}{7}$. Identify correct statement(s)

For More Join: @IITJEE_Advanced

- (A) $\sin^2 3a - \sin^2 a = \sin 2a \sin 3a$
 (B) $\cos a$ is a root of the equation $8x^3 + 4x^2 - 4x - 1 = 0$
 (C) $\cos a$ is irrational
 (D) $\tan 2a + \tan^2 2a + \tan^2 3a = 21$

SECTION-I (iii)

1)

	List-I	List-II	
(P)	$f(x) = \sin\left(\cos\left(e^{\{2x\}}\right)\right) + e^{\cos(5x)}$, is (where $\{.\}$ is fractional part function)	(1)	aperiodic
(Q)	$f(x) = \begin{cases} x - [x], & 2n \leq x < 2n + 1 \\ \frac{1}{2}, & 2n + 1 \leq x < 2n + 2 \end{cases}$; ($n \in \mathbb{I}$), is (where $[.]$ denotes the greatest integer function)	(2)	periodic
(R)	$f(x) = (-1)^{\left[\frac{2x}{\pi}\right]}$, is (where $[.]$ denotes the greatest integer function)	(3)	periodic with fundamental period 2
(S)	$f(x) = ax - [ax + a] + \tan\left(\frac{\pi x}{4}\right)$; ($a \in \mathbb{N}$), is (where $[.]$ denotes the greatest integer function)	(4)	periodic with fundamental period 4
		(5)	periodic but fundamental period can't be determined

- (A) $P \rightarrow 1; Q \rightarrow 2, 3; R \rightarrow 2; S \rightarrow 2, 4$
 (B) $P \rightarrow 3, 4; Q \rightarrow 1, 3; R \rightarrow 2, 4; S \rightarrow 5, 3$
 (C) $P \rightarrow 2, 1; Q \rightarrow 4, 5; R \rightarrow 3, 2; S \rightarrow 2, 3$
 (D) $P \rightarrow 5, 1; Q \rightarrow 4, 2; R \rightarrow 3, 1; S \rightarrow 3, 2$

2)

	List-I	List-II	
(P)	Number of integers in the range of function $f(x) = \sin^2 x + \log_5\left(1 - [\cos^2 x]\right)$ is (Where $[.]$ is greatest integer function)	(1)	1
(Q)	Number of integers in the domain of the function $g(x) = \sin^{-1}([x]) + \tan^{-1}\left(\frac{1}{[x/2]}\right)$ is (Where $[.]$ is greatest integer function)	(2)	2
(R)	Number of real solution(s) of the equation $(2\pi)^{ x } (4 - x) = 1$ is	(3)	4
(S)	Number of real solution(s) of the equation $ \log_3 x + 1 = [x + 1] + [-x + 7]$ is (Where $[.]$ is greatest integer function)	(4)	5

For More Join: @IITJEE_Advanced

		(5)	3
--	--	-----	---

- (A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$
 (B) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$
 (C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$
 (D) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

3)

	List-I	List-II	
(P)	$\frac{8(\sin^3 10^\circ + \sin^3 80^\circ) - \sqrt{4 - 2\sqrt{3}}}{2(\cos 10^\circ + \sin 10^\circ)}$ is equal to	(1)	2
(Q)	Value of $\sum_{r=1}^{2013} \frac{\cos(2 \cdot 3^{r-1} \theta)}{\sin(3^r \theta)}$ where $\theta = 18^\circ$ is equal to	(2)	4
(R)	Minimum value of $8(\cos^2 A + \cos^2 B + \cos^2 C)$ in a triangle ABC, is	(3)	1
(S)	The absolute value of $\frac{1}{2^{98}} \prod_{r=1}^{100} \left(1 - \tan^2 \left(\frac{2^r \pi}{2^{100} + 1}\right)\right)$ is	(4)	6
		(5)	3

- (A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 2$
 (B) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$
 (C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$
 (D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 5$

4)

	List-I	List-II	
(P)	$f(x) = \left[\frac{4x}{\pi} \right]$	(1)	range of $f(x)$ contains atleast one integer but not more than seven
(Q)	$f(x) = \cos^{-1} \left(\operatorname{sgn} \left(\cos \left(\left(\frac{2x-1}{2} \right) \pi \right) \right) \right)$	(2)	is many one in $[-2, 2]$
(R)	$f(x) = \max(\{x+1\}, \{5-x\})$	(3)	is even function in $[-2, 2]$
(S)	$f(x) = \sqrt{x^2} + [x]^2$	(4)	is neither odd nor even function in $[-2, 2]$
		(5)	is bounded function

(Where $[.]$ is greatest integer function ; $\{.\}$ is fractional part function and $\operatorname{sgn}(\cdot)$ is signum function)

- (A) $P \rightarrow 2, 4; Q \rightarrow 1, 2, 4, 5; R \rightarrow 2, 3, 5; S \rightarrow 2, 4$
 (B) $P \rightarrow 1, 2; Q \rightarrow 3, 1, 2; R \rightarrow 4, 3; S \rightarrow 5, 1$
 (C) $P \rightarrow 1, 2, 3, 5; Q \rightarrow 4, 3; R \rightarrow 5, 1; S \rightarrow 3, 1, 2$
 (D) $P \rightarrow 1, 2, 3; Q \rightarrow 3, 4; R \rightarrow 1, 2, 3; S \rightarrow 4, 3, 2$

For More Join: @IITJEE_Advanced

SECTION-II

1) Given that $f(x)$ is a function defined on $(-\infty, \infty)$ satisfying $f(1) = 5$ and for any $x \in (-\infty, \infty)$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = 2f(x) + 9 - 2x$, then $g(2025)$ is

2) If $\sum_{r=0}^{\infty} \tan^{-1} \left(\frac{\cot^{-1}(r^2 + 3r + 3)}{1 + \cot^{-1}(r+1) \cdot \cot^{-1}(r+2)} \right) = \tan^{-1}(a\pi)$, then value of $(2)^{1/a}$ is

3) Number of real solution(s) of the equation

$$11x^2 - 12x + 12\operatorname{cosec}^2(\pi x) + 13\sec^2\left(\frac{9\pi x}{4}\right) + 9\tan^2\left(\frac{9\pi x}{5}\right) = 0 \text{ is}$$

4) Let
$$p = \log_2 \left(\frac{8 \cos\left(\frac{\pi}{6} - 2\theta\right) - \frac{8}{\cot \theta + \cot\left(\frac{\pi}{6} - \theta\right)}}{\sqrt{3}} \right)$$
 (wherever defined) then minimum value of the function $f(x) = ||x-1| + |x-p-1| - |px-1|| + ||x-1| + |x-3| + |px-1||$ is

5) If range of the function $f(x) = \tan^{-1}(3x^2 + (k-1)x + 3)$ is $\left[0, \frac{\pi}{2}\right)$ for $x \in \mathbb{R}$, then the sum of all possible real value(s) of k , is

6) Let $f(x) = \frac{2(4)^{x/2} + 8}{(4)^x + 8(2)^x + 16}$ be a function then the value of $4 \left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \dots + f\left(\frac{59}{15}\right) \right)$ is equal to

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	C	A	B	A

SECTION-I (ii)

Q.	5	6	7
A.	B, C, D	A, B	A, C, D

SECTION-I (iii)

Q.	8	9	10	11
A.	B	A	A	A

SECTION-II

Q.	12	13	14	15	16	17
A.	1	3	2	2	5	3

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	18	19	20	21
A.	A	D	C	C

SECTION-I (ii)

Q.	22	23	24
A.	A, B, D	A, C	A, B

SECTION-I (iii)

Q.	25	26	27	28
A.	D	D	D	B

SECTION-II

Q.	29	30	31	32	33	34
A.	11	205	7	8	23	419

PART-3 : MATHEMATICS

For More Join: @IITJEE_Advanced

SECTION-I (i)

Q.	35	36	37	38
A.	C	A	B	B

SECTION-I (ii)

Q.	39	40	41
A.	B,C	A,B,C	A,B,C,D

SECTION-I (iii)

Q.	42	43	44	45
A.	A	B	B	A

SECTION-II

Q.	46	47	48	49	50	51
A.	17	16	0	4	2	59

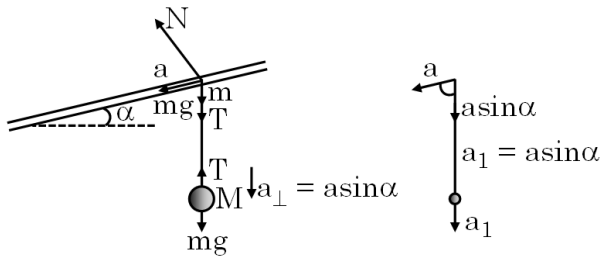
For More Join: @IITJEE_Advanced

SOLUTIONS

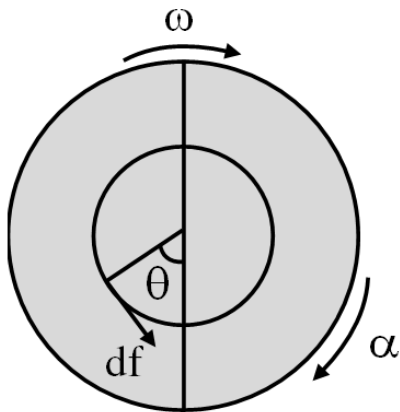
PART-1 : PHYSICS

$$\begin{aligned}
 1) \quad \frac{dU}{dt} &= \frac{1}{2}k \cdot 2x \cdot \frac{dx}{dt} \\
 &= kx \cdot v_{\text{rel}} \\
 &= F_{\text{sp}} \times v_{\text{rel}} \\
 20 &= F_{\text{sp}} \times 2 \\
 \Rightarrow F_{\text{sp}} &= 10 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 2) \quad V \cos \theta &= u \\
 \alpha + \beta + 90^\circ + \theta &= 180^\circ \\
 V &= \frac{u}{\sin(\alpha + \beta)}
 \end{aligned}$$



$$\begin{aligned}
 3) \quad Mg - T &= Ma \sin \alpha \quad \dots(1) \\
 \text{Ring along the Rod} \\
 mg \sin \alpha + T \sin \alpha &= ma \quad \dots(2) \\
 a &= \frac{(m + M)g \sin \alpha}{m + M \sin^2 \alpha} \\
 a_1 &= a \sin \alpha \\
 &= \frac{(m + M)g \sin^2 \alpha}{m + M \sin^2 \alpha}
 \end{aligned}$$



$$\begin{aligned}
 4) \quad df &= \mu dN \\
 df_{y_1} &= \mu \cdot dm g \cdot \sin \theta \\
 &= \mu \cdot \frac{m}{\pi R^2} \cdot g \cdot \sin \theta \cdot dr \cdot r d\theta
 \end{aligned}$$

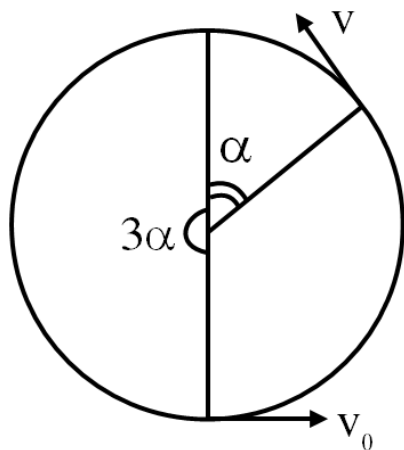
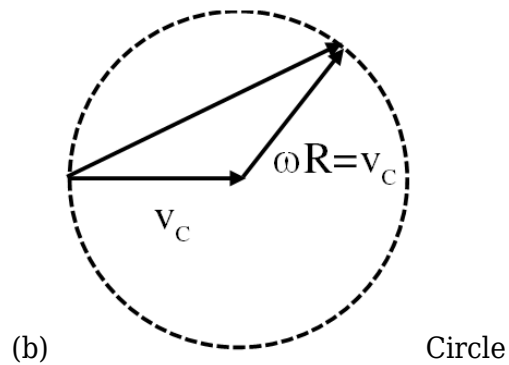
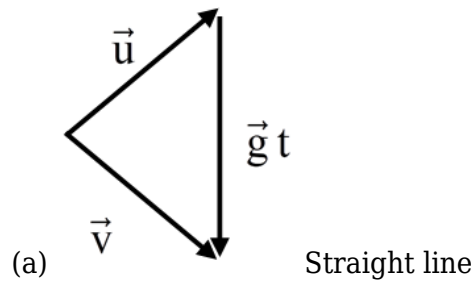
$$f_{y1} = \frac{\mu mg}{\pi R^2} \cdot 2 \cdot \frac{R^2}{2}$$

$$= \frac{\mu mg}{\pi} \text{ (down)}$$

$$f_{y2} = \frac{2\mu mg}{\pi} \text{ (up)}$$

So, $a_{cm} = \frac{\mu g}{\pi} \text{ (up)}$

5)



6)

As $3\alpha = 180^\circ$

$\alpha = 60^\circ$

$$mg \cos \alpha = \frac{mv^2}{R}$$

$$\Rightarrow v = \sqrt{\frac{Rg}{2}}$$

$$\Rightarrow v_0 = \sqrt{\frac{Rg}{2} + 2g \cdot \frac{3R}{2}}$$

For More Join: @IITJEE_Advanced

$$= \sqrt{\frac{7Rg}{2}}$$

Angle of v with horizontal is 60°

$$\text{maximum height} = \frac{R}{2} + \frac{v^2 \sin^2 \alpha}{2g}$$

$$= \frac{R}{2} + \frac{3R}{16}$$

$$= \frac{11R}{16}$$

7)

$$x^2 + h^2 = \ell^2$$

$$x \cdot \frac{dx}{dt} = \ell \cdot \frac{d\ell}{dt}$$

$$xv = \ell v \Rightarrow v = u \sec \alpha.$$

$$xa + v^2 = u^2$$

$$a = \frac{u^2 - v^2}{x} = -\frac{u^2 \tan^2 \alpha}{h \cot \alpha} = -\frac{v^2 \sin^2 \alpha}{h \cot \alpha}$$

$-\alpha$ is increasing & u is constant as a is increasing.

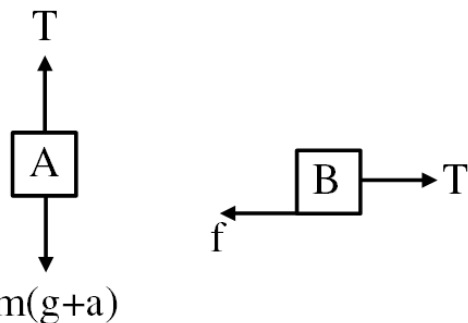
$$T \cos \alpha = ma$$

$$\Rightarrow T = \frac{mv^2 \sin^3 \alpha}{h \cos^2 \alpha}$$

Block is moving on a straight line, so there is no normal acceleration.

8)

In frame of lift



$$f = 0.5 \text{ N}$$

$$1.5 - f = 2 \text{ mar} \quad \text{along the string}$$

$$\Rightarrow a_r = 5 \text{ m/s}^2$$

$$\Rightarrow T = 1 \text{ N}$$

$$a_{Ay} = 0$$

$$a_{Bx} = 5 \text{ m/s}^2$$

$$S_{Ay} = 0$$

$$S_{By} = \frac{1}{2} \times \left(\frac{2}{\sqrt{5}} \right)^2 \times 5 = 2 \text{ m}$$

$$W_A = 0$$

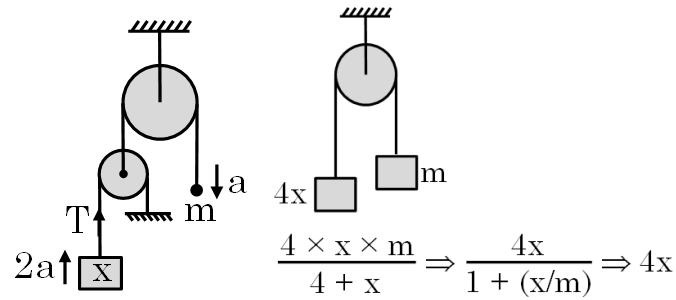
$$W_B = 2 \text{ N}$$

$$9) \frac{4mx}{m+x} = 3x$$

$$4m = 3m + 3x$$

For More Join: @IITJEE_Advanced

$$3x = m$$



$$T - xg = x \times 2a - (1) \times 2$$

$$mg - 2T = ma \dots (2)$$

$$mg - 2xg = (4x + m)a$$

$$a = \frac{mg - 2xg}{4x + m} = \frac{(m - 2x)g}{4x + m}$$

$$\frac{T - xg}{mg - 2T} = \frac{2x}{m}$$

$$mT - mxg = 2mgx - 4Tx$$

$$(m + 4x)T = 3mgx$$

$$T = \frac{3mgx}{m + 4x}$$

$$x = \frac{m}{2}, a = 0$$

$$\text{Case-I : } x = \frac{m}{2}, a = 0$$

$$T = \frac{3mg \times (m/2)}{m + 2m} = \frac{mg}{2}$$

$$a = \frac{mg}{5m} = 2 \text{ m/s}^2$$

$$\text{Case-II : } x = m,$$

$$T = \frac{3mg \times m}{5m} = \frac{3mg}{5}$$

$$x = \frac{m}{4}, a = \frac{m/2g}{2m} = \frac{g}{4} = \frac{10}{4} = 2.5 \text{ m/s}^2$$

$$\text{Case-III : } x = \frac{m}{4}, a = \frac{m/2g}{2m} = \frac{g}{4} = \frac{10}{4} = 2.5 \text{ m/s}^2$$

$$T = \frac{3mg \times (m/4)}{2m} = \left(\frac{3mg}{8} \right)$$

$$\text{Case-IV : } x = 2m$$

$$T = \frac{3mg}{4m} = \frac{2mg}{3}$$

$$a = \frac{3mg}{9m} = \frac{g}{3}$$

$$10) \text{ Diagram of a block of mass } 3m \text{ on an inclined plane of angle } \theta = 45^\circ. \text{ A block of mass } m \text{ is on top of the } 3m \text{ block. The acceleration of the } 3m \text{ block along the incline is } ax' \text{ and the acceleration of the } m \text{ block vertically is } ax. \text{ The normal force is } N. \text{ The distance from the bottom of the } 3m \text{ block to the top of the } m \text{ block is } 2m. \text{ The angle } \theta \text{ is } 45^\circ.$$

$$2a_x = 3a_x \dots (1)$$

$$a_y = a_x + a_x$$

$$a_y = a_x + \frac{3a_x}{2} = \frac{5a_x}{2} \dots (2)$$

$$\text{For } (m + 2m)$$

$$\Rightarrow 3mg - N \cos \theta = 3ma_y \dots (3)$$

For 3m

$$N \sin \theta = 3 m a_x \quad \dots(4)$$

$$\frac{3mg - N \cos \theta}{N \sin \theta} = \frac{5}{2}$$

$$6mg - 2N \cos \theta = 5N \sin \theta$$

$$6mg = \frac{7N}{\sqrt{2}}$$

$$\frac{N}{\sqrt{2}} = \frac{6mg}{7}$$

$$3my - \frac{6mg}{7} = 3ma_y$$

$$a_y = \frac{5g}{7}$$

$$a_x = \frac{2g}{7}$$

$$a'_x = \frac{3g}{7}$$

$$a_{2m} = \sqrt{a_y^2 + a_x^2} = \sqrt{34} \frac{g}{7}$$

11) a_1 = acceleration of m w.r.t to T

a_2 = acceleration of 2m w.r.t to T

a = acceleration of w.r.t to ground

$$\left(\sum f_x \right) = 0$$

$$\Rightarrow m(a + a_1) + 2m(a - a_2) + (m + 5m + 2m)a = 0$$

$$a + a_1 + 2a - 2a_2 + 8a = 0$$

$$11a + a_1 - 2a_2 = 0$$

$$\text{For } m, mg - ma = 2ma_1$$

$$a_1 = \frac{g - a}{2} \quad \dots(2)$$

$$\text{For } 2m, 2mg + 2ma = 4ma_2$$

$$a_2 = \frac{g + a}{2} \quad \dots(3)$$

$$11a + \frac{g - a}{2} - 2 \left(\frac{g + a}{2} \right) = 0$$

$$11a + \frac{g}{2} - \frac{a}{2} - g - a = 0$$

$$19a = g$$

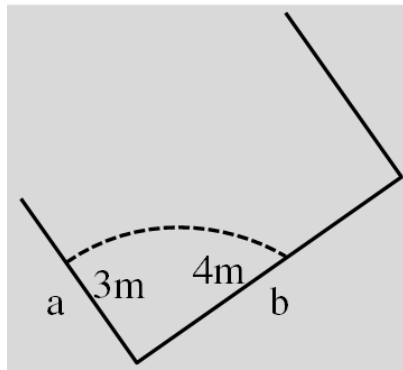
$$a = \frac{g}{19}$$

$$a_1 = \frac{g - a}{2} = \frac{g - (g/19)}{2} = \frac{9g}{19}$$

$$a_2 = \frac{g + a}{2} = \frac{10g}{19}$$

$$a_{12} = a_1 + a_2 = g$$

For More Join: @IITJEE_Advanced



12)

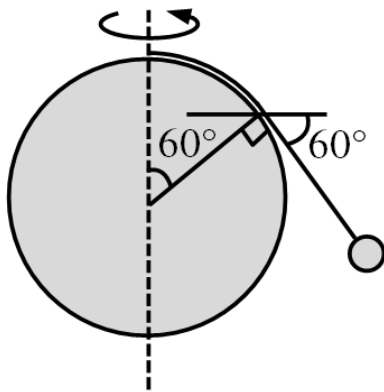
$$a = \frac{1}{2} \cdot g \sin \theta \cdot t^2$$

$$b = \frac{1}{2} \cdot g \cos \theta \cdot t^2$$

$$a^2 + b^2 = \frac{g^2 t^4}{4}$$

$$\Rightarrow t = \left\{ \frac{4(a^2 + b^2)}{g^2} \right\}^{\frac{1}{4}}$$

$$= 1 \text{ sec}$$



13)

$$T \sin 60^\circ = mg$$

$$T \cos 60^\circ = m\omega^2 r$$

$$\Rightarrow \sqrt{3}\omega^2 r = g$$

$$\text{where } r = R \sin 60^\circ + \frac{\pi R}{6} \cdot \cos 60^\circ$$

$$\Rightarrow \sqrt{3} \times 4\pi^2 v^2 \times R \left\{ \frac{\sqrt{3}}{2} + \frac{\pi R}{12} \right\} = g$$

$$\Rightarrow R = \frac{3g}{\sqrt{3}\pi^2 v^2 \cdot (6\sqrt{3} + \pi)}$$

$$= \frac{3g}{\pi^2 v^2 (18 + \sqrt{3}\pi)}$$

14)

$$mg \sin \alpha \cdot x - \mu mg \cos \alpha \cdot x = k$$

$$W_f = - \left(\frac{\mu \cos \alpha}{\sin \alpha + \mu \cos \alpha} \right) k$$

For More Join: @IITJEE_Advanced

$$= -2J$$

$$15) \frac{1}{2} \cdot \frac{mM}{m+M} \cdot u_0^2 = \mu mg \ell$$

$$\Rightarrow \ell = \frac{Mu_0^2}{2\mu g(m+M)} = \frac{u_0^2}{2\mu g(1 + \frac{m}{M})}$$

$$= \frac{1}{2(1 + \frac{1}{4})} = \frac{2}{5}m = 40\text{cm}$$

So the block goes to right end and comes back 15 cm.
It is 10 cm from the left end.

16)

The thread makes angle θ with horizontal, such that.

$$\tan \theta = \frac{h}{2\pi R}$$

Then

$$N = \left\{ (mg \cos \theta)^2 + \left(\frac{mv^2 \cos^2 \theta}{R} \right)^2 \right\}^{\frac{1}{2}}$$

$$mg \sin \theta = \mu \left\{ (mg \cos \theta)^2 + \left(\frac{mv^2 \cos^2 \theta}{R} \right)^2 \right\}^{\frac{1}{2}}$$

$$\text{So, } v = \left\{ \left(\frac{\sin \theta}{\mu} \right)^2 - (\cos \theta)^2 \right\}^{\frac{1}{2}} \cdot \frac{gR}{\cos^2 \theta}$$

$$= \sqrt{5} \text{m/s}$$

$$17) \frac{\mu mg}{k} = e \Rightarrow \mu mg = ek$$

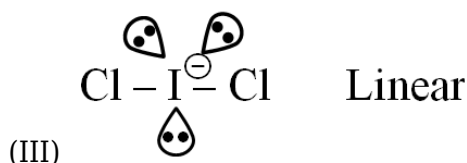
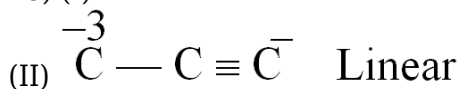
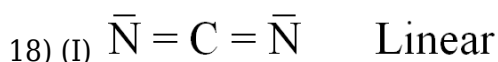
The block oscillates with shifted mean position.

$$\text{Total distance} = (5.5e + 3.5e) + (3.5e + 1.5e) + (1.5e - 0.5e)$$

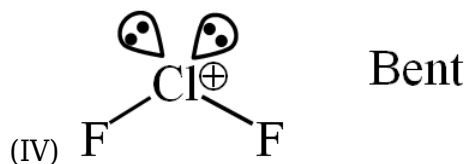
$$= 15e$$

$$\text{So, } W_f = -(15e) \cdot (ek)$$

PART-2 : CHEMISTRY



For More Join: @IITJEE_Advanced



19)

The correct answer is option (D)

20)

The correct answer is option (C)

21)

The correct answer is option (C)

22)

The correct answer is option (A),(B),(D)

23) (A) = All are gases at 25°C

(B) = $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2$

(C) = ΔH_f° HF < HCl < HBr < HI

(D) = B.P. HF > HI > HBr > HCl

24)

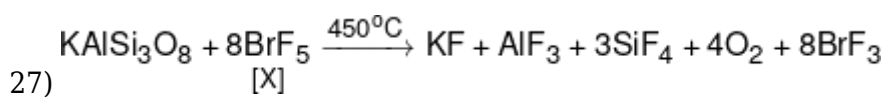
The correct answer is option (A),(B)

25)

The correct answer is option (D)

26)

The correct answer is option (D)



28)

The correct answer is option (B)

For More Join: @IITJEE_Advanced

$$29) 1 = N_2$$

$$2 = HI$$

$$3 = Cl_2$$

$$4 = O_2$$

$$5 = H_2$$

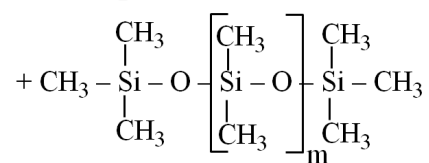
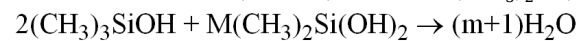
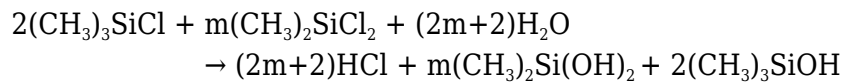
$$6 = CO$$

$$30) P = 50$$

$$Q = 2$$

$$R = 102$$

$$S = 51$$



$$31)$$

The correct answer is 7

$$32)$$

The correct answer is 8

$$33) \text{ The correct answer is 23}$$

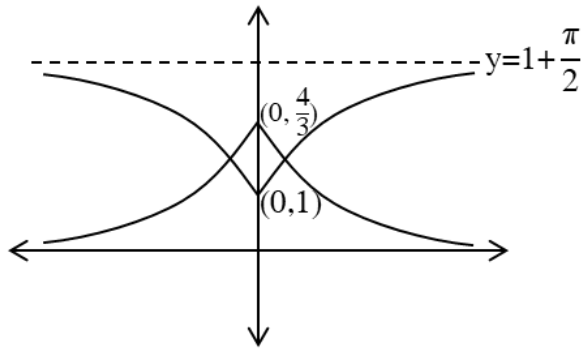
$$34)$$

The correct answer is 419

PART-3 : MATHEMATICS

$$35) \tan^{-1} |x| + 1 = \frac{4}{3} 2^{-|x|}$$

For More Join: @IITJEE_Advanced



36) $\sec^2 x - \tan^2 x = 1$ (on replacing y with x)

$\Rightarrow$ Reflexive

$$\sec^2 x - \tan^2 y = 1$$

$$\Rightarrow 1 + \tan^2 x + 1 - \sec^2 y = 1$$

$$\Rightarrow \sec^2 y - \tan^2 x = 1$$

$\Rightarrow$ symmetric

$$\sec^2 x - \tan^2 y = 1,$$

$$\sec^2 y - \tan^2 z = 1$$

Adding both

$$\Rightarrow \sec^2 x - \tan^2 y + \sec^2 y - \tan^2 z = 1 + 1$$

$$\sec^2 x + 1 - \tan^2 z = 2$$

$$\sec^2 x - \tan^2 z = 1$$

$\Rightarrow$ Transitive

hence equivalence relation

37) Note that $\sin(\alpha + \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta = -\frac{1}{2} + \cos\alpha \sin\beta$.

Because $-1 < \sin(\alpha + \beta) \leq 1$, it follows that $-\frac{1}{2} \leq \cos\alpha \sin\beta < \frac{3}{2}$.

Similarly, because $\sin(\alpha - \beta) = \sin\alpha \cos\beta - \cos\alpha \sin\beta$, we conclude that $-\frac{3}{2} \leq \cos\alpha \sin\beta < \frac{1}{2}$.

Combining the above results shows that $-\frac{1}{2} \leq \cos\alpha \sin\beta \leq \frac{1}{2}$.

38) $\sqrt{\sin(1-x)} = \sqrt{\cos x}$

$$\sin(1-x) = \cos x$$

$$\cos x - \cos\left(\frac{\pi}{2} - (1-x)\right) = 0$$

$$-2 \sin\left(x + \frac{\pi}{4} - \frac{1}{2}\right) \sin\left(\frac{1}{2} - \frac{\pi}{4}\right) = 0$$

$$\sin\left(x + \frac{\pi}{4} - \frac{1}{2}\right) = 0$$

$\square$

$$\square x + \frac{\pi}{4} - \frac{1}{2} = n\pi$$

$$x = \frac{1}{2} - \frac{\pi}{4} \text{ (negative), } x = \frac{1}{2} + \frac{3\pi}{4} \text{ (rejected)}$$

For More Join: @IITJEE_Advanced

$$x = \frac{1}{2} + \frac{7\pi}{4} \text{ (least positive)}$$

$$39) f(x) = \frac{\sqrt{2x^2-1}}{|x|} \quad \& \quad x \in \left[-1, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$$

$$\square f(f(x)) = \frac{\sqrt{2f^2(x)-1}}{|f(x)|} = \frac{\sqrt{3x^2-2}}{\sqrt{2x^2-1}}$$

$$\text{given } f(f(x)) = f(x)$$

$$\square 2x^2 - 1 = |x| \sqrt{3x^2 - 2}$$

$$\text{Let } x^2 = t \Rightarrow 2t - 1 = \sqrt{3t^2 - 2t}$$

$$\Rightarrow t = 1 \text{ or } -1 \text{ (reject)}$$

$$\square x^2 = 1$$

$$\Rightarrow x = 1 \text{ or } -1$$

40)

$$\square g(\alpha+1, \beta) = g(\alpha, \beta) + \alpha$$

$$\begin{aligned} \square g(x, y) &= g(x-1, y) + (x-1) \\ &= g(x-2, y) + (x-2) + (x-1) \\ &= g(x-3, y) + (x-3) + (x-2) + (x-1) \\ &\vdots \\ &\vdots \end{aligned}$$

$$g(x, y) = g(1, y) + \frac{x(x-1)}{2} \quad \dots(1)$$

$$\text{Now, } \square g(\alpha, \beta+1) = g(\alpha, \beta) - \beta$$

$$\begin{aligned} g(1, y) &= g(1, y-1) - (y-1) \\ &= g(1, y-2) - \{(y-2) + (y-1)\} \end{aligned}$$

$$g(1, y) = g(1, 1) - \frac{y(y-1)}{2} \quad \dots(2)$$

from (1) and (2)

$$g(x, y) = 2 - \frac{y(y-1)}{2} + \frac{x(x-1)}{2}$$

$$\Rightarrow (x-y)(x+y-1) = 2 \times 1999$$

$$\Rightarrow (x, y) \equiv (1001, 999)$$

or

$$\equiv (2000, 1999)$$

$$41) \tan^2 a + \tan^2 2a + \tan^2 3a = 21$$

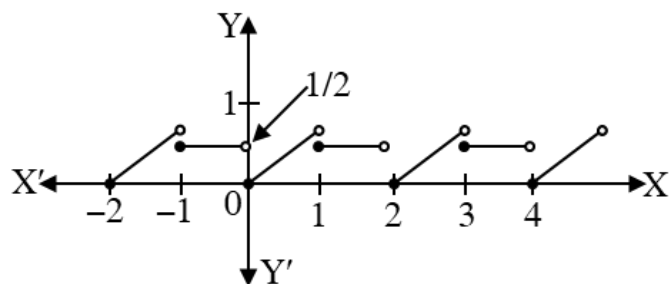
42)

$$(P) \quad f(x) = \begin{cases} 1 & ; x \in \mathbb{Q} \\ 0 & ; x \notin \mathbb{Q} \end{cases}$$

function is periodic but fundamental period can't be determined

For More Join: @IITJEE_Advanced

$$(Q) \quad f(x) = \begin{cases} x - [x] & ; \quad 2n \leq x < 2n + 1 \\ \frac{1}{2} & ; \quad 2n + 1 \leq x < 2n + 2 \end{cases}$$



□ $f(x)$ is periodic with period 2.

$$(R) \quad f(x) = (-1)^{\left[\frac{2x}{\pi}\right]}$$

$$= f(x + \pi) = (-1)^{\left[\frac{2(x+\pi)}{\pi}\right]} = (-1)^{\left[\frac{2x}{\pi}\right]}$$

□ $f(x)$ is periodic with period π

$$(S) \quad f(x) = (ax + a) - [ax + a] + \tan\left(\frac{\pi x}{4}\right)$$

$$= \{ax + a\} + \tan\left(\frac{\pi x}{4}\right)$$

□ period of $f(x)$ is LCM of $\left\{\frac{1}{a}, \frac{4}{1}\right\} = 4$

43)

$$(P) \quad [\cos^2 x] = 0$$

$$\Rightarrow 0 \leq \cos^2 x < 1$$

so, $f(x) = \sin^2 x$

Range $\equiv (0, 1]$

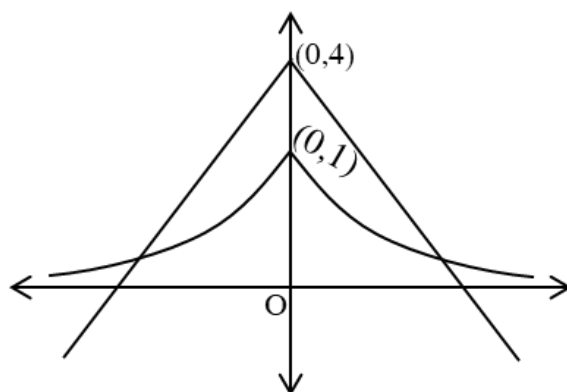
$$(Q) \quad \sin^{-1}[x] \text{ is defined if}$$

$$-1 \leq x < 2$$

but $\left[\frac{x}{2}\right] \neq 0$

Domain $\equiv [-1, 0)$

$$(R) \quad 4 - |x| = (2\pi)^{-|x|}$$



$$(S) \quad \text{if } x \in I$$

$$|\log_3 |x + 1|| = 8$$

$$\Rightarrow |x + 1| = 3^8 \quad \text{or} \quad |x + 1| = 3^{-8}$$

For More Join: @IITJEE_Advanced

↓	↓
Two integral sol. if $x \notin I$ $ \log_3 x+1 = 7$ $\Rightarrow x+1 = 3^7$ ↓ Two integral sol. (rejected)	Two non integral sol. (rejected) $ x+1 = 3^{-7}$ ↓ Two non integral sol.
Number of solutions = 4	

44)

$$\begin{aligned}
 (P) \quad & \frac{8 \{(\sin 10^\circ + \cos 10^\circ)(1 - \sin 10^\circ \cos 10^\circ)\} - 2\sqrt{2} \sin 15^\circ}{2(\sin 10^\circ + \cos 10^\circ)} \\
 &= (4 - 2 \sin 20^\circ) - \frac{2 \sin 15^\circ \cos 45^\circ}{\sin 10^\circ + \cos 10^\circ} \\
 &= 4 - 2 \sin 20^\circ - \frac{(2 \sin 15^\circ \cos 45^\circ)}{2 \sin 45^\circ \cos 35^\circ} \\
 &= 4 - 2 \sin 20^\circ - \frac{\sin 15^\circ}{\cos 35^\circ} \\
 &= 4 - \frac{(\sin 55^\circ - \sin 15^\circ + \sin 15^\circ)}{\cos 35^\circ} \\
 &= 4 - 1 = 3
 \end{aligned}$$

$$\begin{aligned}
 (Q) \quad & 2\alpha + 3\alpha = 90^\circ \\
 & \Rightarrow \cos(2 \cdot 3^{r-1} \alpha) = \cos(90 \cdot 3^{r-1} - 3^r \alpha) \\
 & \text{if } r \text{ is odd then } \cos(2 \cdot 3^{r-1} \alpha) = \sin(3^r \alpha) \\
 & \text{if } r \text{ is even then } \cos(2 \cdot 3^{r-1} \alpha) = -\sin(3^r \alpha)
 \end{aligned}$$

$$\sum_{r=1}^{2013} \frac{\cos(2 \cdot 3^{r-1} \theta)}{\sin(3^r \theta)} = 1$$

$$(R) \quad A = B = C = 60^\circ$$

$$(S) \quad \text{Let } \frac{\pi}{2^{100} + 1} = \theta$$

$$T_r = \frac{\cos(2^{r+1} \theta)}{\cos^2(2^r \theta)}$$

Now

The value of the product is 4

45)

$$(P) \quad f(x) = \left[\frac{4x}{\pi} \right]$$

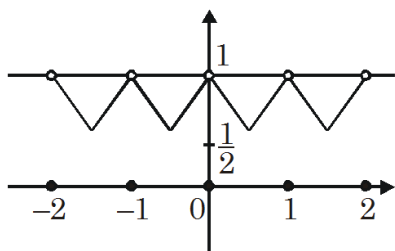
$f(x)$ is many one, is neither odd nor even in $[-2, 2]$

$$(Q) \quad f(x) = \max(\{x\}, \{-x\})$$

range contains at least one integer but not more than 3 integers.

$f(x)$ is many one, bounded and neither odd nor even.

For More Join: @IITJEE_Advanced



(R)

(S) $f(x) = |x| + [x]^2$
is many one and neither odd nor even

46)

$$f(x+1) \leq f(x) + 1$$

$$\Rightarrow f(x+5) \leq f(x+4) + 1$$

$$\leq f(x+3) + 2$$

$$\leq f(x+2) + 3$$

$$\leq f(x+1) + 4$$

$$\Rightarrow f(x+5) \leq f(x) + 5 \quad \dots(1)$$

and given

$$f(x+5) \geq f(x) + 5 \quad \dots(2)$$

equality must be held everywhere

$f(1), f(2), f(3), \dots$ A.P.

$\Rightarrow 5, 6, 7, \dots$

$$f(2025) = 2029$$

$$g(2025) = 2f(2025) + 9 - 2(2025)$$

$$g(2025) = 2017$$

47)

$$\begin{aligned} T_r &= \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{(r+2)-(r+1)}{1+(r+2)(r+1)} \right)}{1 + \tan^{-1} \left(\frac{1}{r+1} \right) \cdot \tan^{-1} \left(\frac{1}{r+2} \right)} \right) \\ T_r &= \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{\frac{1}{r+1} - \frac{1}{r+2}}{1 + \frac{1}{r+1} \cdot \frac{1}{r+2}} \right)}{1 + \tan^{-1} \left(\frac{1}{r+1} \right) \cdot \tan^{-1} \left(\frac{1}{r+2} \right)} \right) \\ &= \tan^{-1} \left(\frac{\tan^{-1} \left(\frac{1}{r+1} \right) - \tan^{-1} \left(\frac{1}{r+2} \right)}{1 + \tan^{-1} \left(\frac{1}{r+1} \right) \cdot \tan^{-1} \left(\frac{1}{r+2} \right)} \right) \\ &= \tan^{-1} \left(\tan^{-1} \left(\frac{1}{r+1} \right) \right) - \tan^{-1} \left(\tan^{-1} \left(\frac{1}{r+2} \right) \right) \\ \sum_{r=0}^{\infty} T_r &= \tan^{-1} \left(\tan^{-1} (1) \right) = \tan^{-1} \left(\frac{\pi}{4} \right) \\ \Rightarrow a &= \frac{1}{4} \Rightarrow 2^{1/a} = 16 \end{aligned}$$

For More Join: @IITJEE_Advanced

$$48) 2x^2 + (3x - 2)^2 + 8 + 12\cot^2(\pi x) + 13\sec^2\left(\frac{9\pi x}{4}\right) + 9\tan^2\left(\frac{9\pi x}{5}\right) = 0$$

$x \in \phi$

49)

$$Q = \frac{8\cos\left(\frac{\pi}{6} - 2\theta\right) - \frac{8}{\cot\theta + \cot\left(\frac{\pi}{6} - \theta\right)}}{\sqrt{3}}$$

Let

$$= \frac{8\cos\left(\frac{\pi}{6} - 2\theta\right) - \frac{8\sin\theta\sin\left(\frac{\pi}{6} - \theta\right)}{\sin\frac{\pi}{6}}}{\sqrt{3}}$$

$$= \frac{8}{\sqrt{3}} \left\{ \cos\left(\frac{\pi}{6} - 2\theta\right) - \cos\left(\frac{\pi}{6} - 2\theta\right) + \cos\left(\frac{\pi}{6}\right) \right\}$$

$\Rightarrow Q = 4$

$$p = 2$$

$$f(x) = ||x - 1| + |x - 3| - |2x - 1|| + ||x - 1| + |x - 3| + |2x - 1||$$

$$f(x) = 2\max\{|x - 1| + |x - 3|, |2x - 1|\}$$

min value of $f(x)$ is 4

$$50) \text{ Let } g(x) = 3x^2 + (k - 1)x + 3; x \in \mathbb{R}$$

$$\square \text{ Range of } g(x) \in [0, \infty)$$

$$\text{so, } D = 0 \Rightarrow (k - 1)^2 - 4 \cdot 3 \cdot 3 = 0$$

$$k - 1 = \pm 6$$

$$\square k = 7, -5$$

$$51) f(x) = \frac{42^x + 16}{2 \cdot 2^{2x} + 16 \cdot 2^x + 32}$$

$$f(x) = \frac{2^{2x} + 8 \cdot 2^x + 16}{2}$$

$$f(x) = \frac{2^x + 4}{2^x}$$

$$f(4 - x) = \frac{1}{2(2^x + 4)}$$

$$f(x) + f(4 - x) = \frac{1}{2}$$

Aliter :

$$f(x) = \frac{2(4)^{x/2} + 8}{(4)^x + 8(2)^x + 16}$$

$$f(x) = \frac{2(2)^x + 8}{(2)^{2x} + 8(2)^x + 16}$$

For More Join: @IITJEE_Advanced

$$\begin{aligned}
 f(4-x) &= \frac{2(2)^{4-x} + 8}{(2)^{8-2x} + 8(2)^{4-x} + 16} \\
 &= \frac{\frac{32}{2^x} + 8}{\frac{2^8}{2^{2x}} + \frac{2^7}{2^x} + 16} \\
 &= \frac{4 + 2^x}{\left(\frac{32}{2^{2x}} + \frac{16}{2^x} + 2\right) 2^x} \\
 f(4-x) &= \frac{2^{2x} + 4(2)^x}{2(2)^{2x} + 16(2)^x + 32} \\
 f(x) + f(4-x) &= \frac{2^{2x} + 4(2)^x + 4(2)^x + 16}{2(2)^{2x} + 16(2)^x + 32} \\
 &= \frac{1}{2} \\
 \Rightarrow f(x) + f(4-x) &= \frac{1}{2}
 \end{aligned}$$

Now

$$\begin{aligned}
 &4 \left\{ \underbrace{\left(\frac{1}{2} + \frac{1}{2} + \dots + \frac{1}{2} \right)}_{29 \text{ times}} + f(2) \right\} \\
 &= 58 + 1 = 59
 \end{aligned}$$

For More Join: @IITJEE_Advanced